

Probabilistic Methods

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Lecture Notes

A clear route from counting and events to random variables and data.

June 8, 2026

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Chapter 1

Introduction

These notes are an introduction to probabilistic thinking: the mathematical art of reasoning under uncertainty. Probability is not presented here as a collection of ready-made formulas, but as a language for describing experiments, incomplete information, variability, randomness, and data.

The purpose of the course is to teach how to build probabilistic models carefully. Before one computes a probability, expectation, variance, confidence interval, or approximation, one should understand what is random, what is being observed, what assumptions are being made, and what mathematical object represents the situation. In this sense, probability is not only a computational tool, but also a discipline of precise modelling.

The course is intended for readers who want to understand why probabilistic methods work, not only how to apply them. The emphasis is therefore placed on interpretation, model construction, and the connection between formal definitions and concrete situations. Calculations appear throughout the notes, but they are treated as consequences of a well-defined model rather than as isolated procedures.

1.1 How to Use These Notes

These notes are meant to be read before problem classes. Their role is not only to present the material, but also to show how mathematical argumentation is built: how one starts from a situation, identifies the relevant objects, formulates assumptions, chooses notation, develops a line of reasoning, and only then performs computations. A problem class is therefore not the place where the basic language of the topic is introduced for the first time; it is the place where that language is used, tested, discussed, and refined through problems.

Students are expected to read the relevant chapter in advance and to treat the text as a model of reasoning. The important question is not only “What is the answer?” but also “How is the solution argued?” While reading, students should pay attention to the order of the argument, the meaning of each definition, the reason for each assumption, and the way examples are connected to general concepts. The aim is to learn how to construct a probabilistic explanation, not merely how to recognize a formula that seems to fit a task.

Problem lists will accompany the chapters and will be used during classes. They are meant to be solved after the corresponding text has been read. The chapter gives the language, concepts, and patterns of argumentation needed to approach the problems responsibly. Without this preparation, a solution may look complete on paper while still being mathematically empty: it may contain formulas, calculations, or generated text, but no clear understanding of what model is being used or why the reasoning is valid.

Artificial intelligence tools may be useful for practice, checking calculations, generating additional examples, or asking for alternative explanations. Students are encouraged to use such tools responsibly when they support learning. However, AI-generated text cannot replace the student’s own reading and understanding. A student must be able to read, question, correct, and explain any solution they present. If one cannot explain the experiment, the elementary outcomes, the assumptions, the chosen model, and the steps of the argument, then the work has not yet been done.

These notes also help students learn what to ask for. To use problem lists, textbooks, or AI tools well, one must first know what matters: which object is being modelled, which assumptions

are decisive, which definitions are involved, and what kind of justification is expected. The PDF is therefore an essential part of the course: it teaches the material, but it also teaches how to read mathematics, how to demand proper explanations, and how to recognize whether an answer contains real reasoning.

The expected preparation for classes is conceptual as well as computational. After reading a chapter, a student should be able to summarize its main ideas, explain the definitions in their own words, reconstruct the key examples, identify the assumptions in a probabilistic model, follow and reproduce the main arguments, and answer questions that test understanding of the text. The notes are deliberately written with full explanations and detailed argumentation so that this preparation is possible.

1.2 Structure of the Course

These lecture notes are organized as a gradual construction of probabilistic thinking. The aim is not to begin with formulas and then search for examples, but to develop the mathematical language needed to describe uncertainty in a precise way. We start from the problem of counting and representing possible outcomes, then move toward events, probability, random variables, probability laws, empirical data, sampling, limit theorems, and statistical inference.

The course begins with combinatorics because many probabilistic models depend on the ability to count possible outcomes correctly. At this stage probability itself is not yet needed. What is needed is the more elementary skill of recognizing what kind of object an outcome is: a sequence, a selection, an arrangement, a choice with or without repetition, or an object whose elements may or may not be distinguishable. This first step builds the counting language that later allows us to describe sample spaces and favorable outcomes with precision.

The second stage has a special conceptual role. It is devoted to discovering the formal language of probability theory. The purpose is to show how a mathematical theory can be built from careful thinking, analysis, and abstraction. We examine how statements about a random situation select sets of outcomes, how these sets become events, how logical operations correspond to set operations, and why a numerical theory of probability should be placed on such a structure. This part is not primarily computational; it is meant to teach how probabilistic language is formed.

After this conceptual preparation, the notes turn to sample spaces and elementary probability. Several canonical examples are developed from the beginning: one first understands the experiment, decides what should count as an elementary outcome, constructs the sample space, defines events, and only then assigns probabilities. Different representations of the same structure, such as lists, tables, trees, and symbolic notation, are used to show that a sample space is an abstract mathematical model rather than a collection of observed data.

The next step is conditional probability. Here the central idea is that information changes the space in which we reason. Conditional probability is introduced by asking what it means to restrict attention to the cases in which some event has already occurred. From this point of view, the formula for conditional probability becomes natural, and it leads to independence, dependence, partitions, the law of total probability, and Bayes' theorem. Bayes' theorem is treated as a central tool for reversing conditional reasoning and for understanding situations where intuition is easily misled.

The course then introduces random variables through the discrete case. A random variable appears as a function that extracts a numerical quantity from an elementary outcome: the number of successes, the number of errors, the number of arrivals, a payoff, or another measurable feature of the experiment. This leads to the support of a discrete random variable, its probability mass function, its cumulative distribution function, expectation, variance, moments, and interpretation. Classical discrete laws are presented as models arising from specific mechanisms, not as isolated formulas.

The continuous case is then developed as a parallel but conceptually more demanding part of the theory. For a continuous random variable, probability is no longer concentrated at individual points, so probabilities of intervals become the fundamental objects. This leads to density functions, cumulative distribution functions, integrals, continuous expectations, variances, moments, quantiles, and the interpretation of probability as area under a density. Important continuous models, including the uniform, exponential, normal, gamma, and beta distributions, are introduced with attention to their meaning and use.

Once theoretical probability laws have been introduced, the course moves to empirical distributions and descriptive statistics. The same ideas now appear in data: frequency tables, histograms, empirical cumulative distribution functions, means, medians, variances, standard deviations, quartiles, interquartile ranges, and outliers. This stage is meant to clarify the difference between an idealized probability law and a finite dataset, and also to show how empirical summaries mirror the theoretical quantities defined earlier.

The following stage adds another level of randomness by studying sampling and distributions of statistics. A sample mean, sample proportion, sample variance, median, or other statistic is not merely a number; it is itself a random variable because it depends on the random sample. This perspective prepares the ground for understanding sampling distributions and for seeing why repeated sampling creates a new layer of probabilistic structure.

The final stage of the present course outline is devoted to limit theorems and statistical inference. Once empirical summaries and sampling distributions are understood, we can ask what regularities appear as the sample size grows or as experiments are repeated many times. This leads to the law of large numbers, the central limit theorem, normal approximations, standard errors, confidence intervals, and the basic logic of statistical inference. The goal is to show how probability theory makes inference from data possible: it allows us to estimate unknown quantities, measure uncertainty, and justify conclusions mathematically.

Chapter 2

Elements of Combinatorics

2.1 Counting begins with elementary outcomes

Combinatorics is often presented as a list of formulas. This is misleading. Before a formula can be used, we must know what one elementary outcome is. A physical situation does not determine a mathematical space of possibilities by itself. The space of possibilities depends on what is recorded.

Suppose that three balls are drawn from a box. If the balls are labelled and the order of drawing is recorded, then an outcome is a sequence of labelled objects. If only the selected labelled balls are recorded, then an outcome is a set. If only the colors are recorded, then labelled balls of the same color may become indistinguishable. If only the number of red balls is recorded, then the outcome is a numerical summary of the draw rather than the full draw itself.

Thus the first question in a counting problem is not

Which formula should be used?

The first question is

What is one outcome?

Definition

An **elementary outcome** is a complete description of one possible result of a situation, according to the recording rule chosen for the model.

The phrase “according to the recording rule” is essential. The same physical situation may produce different mathematical spaces of possibilities if different information is recorded.

Example

A box contains three balls labelled R_1, R_2, B_1 . Two balls are drawn without replacement. If order and labels are recorded, then examples of outcomes are

$$(R_1, B_1), \quad (B_1, R_1), \quad (R_1, R_2).$$

If only the set of labelled balls is recorded, then examples of outcomes are

$$\{R_1, B_1\}, \quad \{R_1, R_2\}.$$

If only the sequence of colors is recorded, then examples of outcomes are

$$(R, B), \quad (B, R), \quad (R, R).$$

If only the number of red balls is recorded, then the possible recorded values are

$$0, 1, 2.$$

These are not four different answers to the same counting question. They are four different mathematical models of the same physical situation.

Remark

A counting error often begins before any arithmetic is done. It begins when the outcome type is chosen incorrectly.

2.2 Recording rules and spaces of possibilities

A recording rule tells us which differences between physical results are mathematically visible. Some differences are recorded; others are ignored. Once the recording rule is fixed, the space of possibilities is the set of all possible recorded outcomes.

Definition

A **recording rule** is a decision about what information is included in the mathematical outcome and what information is ignored.

For finite situations, the construction usually follows this pattern:

physical situation $\longrightarrow$ recording rule $\longrightarrow$ elementary outcome $\longrightarrow$ space of possibilities.

The following table shows common outcome types.

Recording rule	Outcome type	Typical notation
Order is recorded	sequence	$(a_1, a_2, \dots, a_k)$
Only membership is recorded	set	$\{a_1, a_2, \dots, a_k\}$
All objects are arranged in a row	linear arrangement	$(a_{\sigma(1)}, \dots, a_{\sigma(n)})$
Objects are arranged on a circle	circular arrangement	relative order
Only numbers of types are recorded	count vector	$(n_1, n_2, \dots, n_r)$
Only a numerical summary is recorded	summary value	x

2.3 Representing the space of possibilities

Counting is easier when the space of possibilities is represented in a form that matches the structure of the problem. A representation is not decoration. It is part of the reasoning, because it makes visible which outcomes are different, which descriptions are equivalent, and which construction steps are being used.

Different situations call for different representations:

- a list is useful when the space is small and every outcome can be written explicitly;
- a table is useful when two choices are made and rows and columns organize the possibilities;
- a tree is useful when an outcome is built through successive stages;
- a set notation is useful when only membership matters;
- a tuple notation is useful when positions or order matter;
- a count vector is useful when only the numbers of objects of each type matter.

Example

If two letters are chosen from $\{A, B, C\}$ and order is recorded with repetition allowed, the space can be represented as a table:

	A	B	C
A	(A, A)	(A, B)	(A, C)
B	(B, A)	(B, B)	(B, C)
C	(C, A)	(C, B)	(C, C)

The table shows two things at once: there are 3 choices for the first position and 3 choices for the second position, and the ordered pairs (A, B) and (B, A) are different outcomes.

Example

If two letters are chosen from $\{A, B, C\}$ and only the selected set is recorded, then

$$\{A, B\} = \{B, A\}.$$

A table of ordered pairs would still be possible as an auxiliary representation, but it would overdescribe the outcomes. The actual outcomes are sets:

$$\{A, B\}, \quad \{A, C\}, \quad \{B, C\}.$$

The change of representation reflects a change of model.

A useful representation should answer the question: what differences are visible in this model? If the representation hides an important difference, the count may be too small. If it records irrelevant differences, the count may be too large.

Example

A committee of three people is chosen from ten people. If the committee is recorded only as a group, then

$$\{A, B, C\} = \{C, A, B\}.$$

The order of listing the names is irrelevant. The outcome is a set.

If the same three people are assigned the roles of chair, secretary, and treasurer, then

$$(A, B, C) \neq (C, A, B).$$

The outcome is now an ordered assignment of roles.

The two situations may look similar in everyday language: in both cases three people are selected. But mathematically they are different because the recording rules are different.

2.4 Diagnostic questions before counting

Before using a formula, answer the following questions.

1. What is one elementary outcome?
2. Is the outcome a sequence, a set, an arrangement, or a count vector?
3. Does order matter?

4. Are all objects used, or only some of them?
5. Can the same object be used more than once?
6. Are the objects distinguishable or indistinguishable?
7. Are we first counting labelled descriptions and then identifying descriptions that represent the same outcome?

These questions are more important than the names of formulas. The names are only labels attached to structures that have already been recognized.

Remark

Do not begin a problem by asking which formula you remember. Begin by asking what kind of outcome is being constructed.

2.5 The multiplication principle as a construction rule

Many counting problems can be solved by describing how an outcome is constructed step by step.

Theorem

Multiplication principle. Suppose a construction is performed in k successive stages. If there are m_1 choices for the first stage, m_2 choices for the second stage, and so on up to m_k choices for the k -th stage, then the number of possible complete constructions is

$$m_1 m_2 \cdots m_k.$$

The multiplication principle is not only a formula. It is a way of reading an outcome as a construction process.

Example

A four-digit PIN code is formed from the digits $0, 1, \dots, 9$, and digits may repeat. One outcome is a sequence such as

$$(0, 7, 7, 3).$$

There are 10 choices for each position, hence

$$10 \cdot 10 \cdot 10 \cdot 10 = 10^4$$

possible PIN codes.

The reason is not that the formula n^k was memorized. The reason is that a PIN code is a sequence of four independent choices from a set of ten symbols.

Example

A three-letter code is formed from the letters A, B, C, D , and letters may not repeat. One outcome is an ordered triple, for example

$$(A, D, B).$$

There are 4 choices for the first position, 3 for the second, and 2 for the third. Therefore

the number of codes is

$$4 \cdot 3 \cdot 2 = 24.$$

When the same number of choices is available at every stage, powers appear. When the number of choices decreases, falling products appear. When order is later ignored, division appears. Most elementary combinatorial formulas can be understood from these three ideas.

2.6 Sequences with repetition

A sequence with repetition is an ordered list in which each position is filled separately from the same set of symbols.

Definition

Let S be a set with n elements. A **sequence of length k with repetition** over S is an ordered k -tuple

$$(a_1, a_2, \dots, a_k),$$

where each $a_i \in S$. The same element may occur many times.

The word “ordered” means that positions matter. The first position, the second position, and the third position are different places in the description. The phrase “with repetition” means that choosing one symbol at one position does not remove it from later positions.

Now we derive the count. To construct one sequence,

- choose a_1 : there are n choices;
- choose a_2 : there are again n choices, because repetition is allowed;
- continue in the same way up to a_k : there are n choices at each position.

By the multiplication principle, the number of complete sequences is

$$\underbrace{n \cdot n \cdots n}_{k \text{ factors}} = n^k.$$

Thus the formula n^k is not a separate rule to memorize. It is the multiplication principle applied to k ordered positions, each with the same n choices.

This model is used when:

- order is recorded,
- each position is chosen separately,
- repetition is allowed,
- the outcome is a full sequence.

Typical examples include PIN codes, repeated coin tosses, repeated die rolls, strings over an alphabet, and repeated selections with replacement.

Example

The space of possibilities for three tosses of a coin is

$$\{H, T\}^3.$$

It has

$$2^3 = 8$$

elementary outcomes. Each outcome is a sequence, for example

$$(H, T, H).$$

The outcomes (H, T, H) and (T, H, H) are different because the order of positions is recorded.

A tree representation shows the same count visually. At the first toss there are 2 branches, from each of these there are 2 branches for the second toss, and from each of those there are 2 branches for the third toss. The tree has

$$2 \cdot 2 \cdot 2 = 8$$

terminal branches, one for each complete sequence.

2.7 Ordered selections without repetition

Sometimes only some objects are chosen, but the order or position of the chosen objects matters.

Definition

An **ordered selection without repetition** of length k from n distinct objects is an ordered list of k different objects chosen from the n available objects.

To derive the count, construct the ordered list position by position. There are n choices for the first position. Once the first object has been used, it cannot be used again, so there are $n - 1$ choices for the second position. After two objects have been used, there are $n - 2$ choices for the third position. Continuing in this way, the k -th position has

$$n - (k - 1) = n - k + 1$$

choices. Therefore the number of ordered selections without repetition is

$$P(n, k) = n(n - 1)(n - 2) \cdots (n - k + 1).$$

This product can also be written using factorials. Since

$$n! = n(n - 1) \cdots (n - k + 1)(n - k)(n - k - 1) \cdots 1,$$

dividing by $(n - k)!$ removes the unused tail:

$$P(n, k) = \frac{n!}{(n - k)!}.$$

This model is used when:

- only some objects are selected,
- order or position matters,
- no object may be used twice.

Example

Gold, silver, and bronze medals are assigned among 12 runners. An outcome is not a set of three runners. It is an ordered triple

$$(\text{gold}, \text{silver}, \text{bronze}).$$

There are 12 choices for gold, then 11 choices for silver, then 10 choices for bronze. Hence the number of possible assignments is

$$12 \cdot 11 \cdot 10 = P(12, 3).$$

In some European terminology, ordered selections are called variations. In these notes we use the name ordered selections without repetition, or k -permutations, because it describes directly what the outcome is.

2.8 Arrangements of all objects

A permutation is an arrangement of all objects in a row.

Definition

A **permutation** of n distinct objects is a linear arrangement of all n objects.

This is the special case of an ordered selection without repetition in which all objects are used. Thus $k = n$, and

$$P(n, n) = n(n-1)(n-2) \cdots 2 \cdot 1 = n!.$$

So there are

$$n!$$

permutations of n distinct objects.

Example

If 7 students are arranged in a line, then one outcome is a complete ordered list of all students. The number of arrangements is

$$7!.$$

This is not because $7!$ is a name attached to the word “arrangement”. It is because there are 7 choices for the first position, 6 for the second, 5 for the third, and so on until all positions are filled.

Permutations are used when:

- all objects are used,
- order matters,
- the objects are distinguishable,
- the arrangement is linear.

2.9 Circular arrangements

A circular arrangement is different from a linear arrangement because rotating the whole arrangement does not create a new relative seating order.

Example

Suppose four people A, B, C, D sit around a round table. The linear lists

$$(A, B, C, D), \quad (B, C, D, A), \quad (C, D, A, B), \quad (D, A, B, C)$$

represent the same circular arrangement if only relative positions matter.

For n distinct objects placed around a circle, with rotations identified, the number of circular arrangements is

$$(n - 1)!$$

One way to see this is to fix one object as a reference point. This does not reduce any real choice; it only chooses one representative from each group of rotated descriptions. Once the reference object is fixed, the remaining $n - 1$ objects can be placed around it in a linear order:

$$(n - 1)(n - 2) \cdots 2 \cdot 1 = (n - 1)!$$

Remark

A circular arrangement requires a modelling decision. If seats are numbered or if there is a distinguished position, then rotations may become different outcomes and the linear count may be appropriate.

2.10 Unordered selections: combinations

A combination is an unordered selection.

Definition

A **combination** of size k from an n -element set is a subset with k elements.

The number of combinations is

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

This formula can be derived from ordered selections. There are

$$P(n, k) = \frac{n!}{(n - k)!}$$

ordered selections of length k . But each unordered selection of k objects can be ordered in $k!$ ways. Therefore

$$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{k!(n - k)!}.$$

Combinations are used when:

- only membership is recorded,
- order is ignored,

- each object may be chosen at most once,
- the outcome is a set.

Example

A five-card hand from a standard deck is a set of five cards, not a sequence of five draws, if the final hand is all that is recorded. The number of possible hands is

$$\binom{52}{5}.$$

If the order of drawing is recorded, the correct count is instead

$$P(52, 5) = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

These are different sample spaces.

2.11 Overcounting and equivalent descriptions

Many combinatorial formulas arise from a two-step idea:

1. Count descriptions that are easy to construct.
2. Divide by the number of descriptions representing the same outcome.

This is the idea of overcounting.

Definition

Overcounting occurs when the same intended outcome is counted more than once because it has several different descriptions.

Overcounting is not a mistake if it is controlled. It becomes a method when we know exactly how many times each outcome has been counted.

Example

Suppose we want to choose a committee of three people from n people. The order of names in the committee does not matter.

It is easier first to count ordered selections:

$$n(n-1)(n-2).$$

But each committee of three people has been counted $3!$ times, because the same three people can be listed in $3!$ different orders. Therefore the number of committees is

$$\frac{n(n-1)(n-2)}{3!}.$$

The essential idea is not division itself. The essential idea is identifying which descriptions should be treated as the same outcome.

2.12 Distinguishable and indistinguishable objects

Objects are distinguishable if the model treats them as separate individuals. Objects are indistinguishable if only their type matters.

Example

The objects

$$R_1, R_2, R_3$$

are three distinguishable red balls. The symbols

$$R, R, R$$

represent three indistinguishable red balls of the same type.

A physical object may be distinguishable in one model and indistinguishable in another. This depends on the recording rule.

Example

An urn contains red and blue balls. If each ball has a label, and labels are recorded, then two red balls are different outcomes. If only colors are recorded, then the red balls are treated as indistinguishable.

This distinction is crucial in probability. If the physical random mechanism selects labelled balls uniformly, but the sample space records only colors, then the color outcomes may not be equally likely.

2.13 Permutations with repeated elements

When some objects are indistinguishable, a linear arrangement may have several labelled descriptions that represent the same visible arrangement.

Suppose we arrange n objects, where there are n_1 objects of type 1, n_2 objects of type 2, and so on, with

$$n_1 + n_2 + \cdots + n_r = n.$$

The number of distinct visible arrangements is

$$\frac{n!}{n_1!n_2!\cdots n_r!}.$$

Example

The word BANANA contains six letters: three A 's, two N 's, and one B . If all six positions are arranged, then the number of distinct words is

$$\frac{6!}{3!2!1!}.$$

The division appears because permuting the three A 's among themselves does not change the visible word, and neither does permuting the two N 's.

This formula should be read as an overcounting correction:

$$\text{labelled arrangements} \longrightarrow \text{visible arrangements.}$$

2.14 Combinations with repetition

Sometimes we choose k objects from n types, and the same type may be chosen several times. The order of the chosen objects is ignored. The outcome is then not a sequence of choices, but a vector of counts.

Definition

A **combination with repetition** of size k from n types is a choice of non-negative integers

$$(x_1, x_2, \dots, x_n)$$

such that

$$x_1 + x_2 + \dots + x_n = k.$$

Here x_i records how many objects of type i are chosen.

The important modelling decision is that the positions of the chosen objects are not recorded. For example, choosing two objects of type 1, none of type 2, and three of type 3 is recorded as

$$(2, 0, 3),$$

not as an ordered sequence of five choices.

The stars and bars representation

We derive the formula by representing each count vector visually. Write k stars to represent the chosen objects and insert $n - 1$ bars to divide the stars into n groups. The number of stars before the first bar is x_1 , the number of stars between the first and second bar is x_2 , and so on, until the number of stars after the last bar is x_n .

For example, if $n = 3$ and $k = 5$, the count vector

$$(2, 0, 3)$$

is represented by

$$** || ***.$$

There are two stars before the first bar, no stars between the two bars, and three stars after the second bar. This represents

$$x_1 = 2, \quad x_2 = 0, \quad x_3 = 3.$$

Conversely, every arrangement of k stars and $n - 1$ bars determines exactly one count vector. For instance,

$$| *** | **$$

represents

$$(0, 3, 2).$$

This is a one-to-one correspondence:

count vectors $(x_1, \dots, x_n)$ with $x_1 + \dots + x_n = k \longleftrightarrow$ arrangements of k stars and $n - 1$ bars.

Therefore the counting problem becomes a simpler arrangement problem. We have

$$k + (n - 1) = n + k - 1$$

total positions. To form an arrangement, we only need to decide which k of these positions contain stars; the remaining $n - 1$ positions will contain bars. Hence the number of arrangements is

$$\binom{n + k - 1}{k}.$$

Equivalently, we may choose the $n - 1$ positions of the bars, giving

$$\binom{n + k - 1}{n - 1}.$$

Thus

$$\binom{n + k - 1}{k} = \binom{n + k - 1}{n - 1}.$$

Theorem

The number of combinations with repetition of size k from n types is

$$\binom{n + k - 1}{k} = \binom{n + k - 1}{n - 1}.$$

Example

How many ways are there to choose 5 donuts from 3 flavors if only the numbers of each flavor matter?

An outcome is a count vector

$$(x_1, x_2, x_3), \quad x_1 + x_2 + x_3 = 5.$$

The stars and bars representation uses 5 stars and 2 bars, so there are 7 positions in total. We may choose the 5 positions of the stars:

$$\binom{7}{5},$$

or the 2 positions of the bars:

$$\binom{7}{2}.$$

Thus the number of possible choices is

$$\binom{3 + 5 - 1}{5} = \binom{7}{5} = \binom{7}{2}.$$

Remark

Combinations with repetition do not describe ordered repeated choices. If order matters, the model is a sequence with repetition and the count is n^k . If order is ignored and only counts of types matter, the model is combinations with repetition.

2.15 Recognizing the correct counting model

The following diagnostic table summarizes the main finite counting structures.

Outcome type	Order?	Repetition?	Count
sequence with repetition	yes	yes	n^k
ordered selection	yes	no	$P(n, k) = \frac{n!}{(n-k)!}$
permutation	yes	no, all used	$n!$
circular arrangement	relative order	no, all used	$(n-1)!$
combination	no	no	$\binom{n}{k}$
combination with repetition	no	yes	$\binom{n+k-1}{k}$
multiset permutation	positions yes	identical objects	$\frac{n!}{n_1! \cdots n_r!}$

The table should be used only after the outcome type has been identified. It is a summary, not a substitute for modelling.

Remark

A formula is a compressed argument. If the argument does not match the situation, the formula gives the wrong count even when the arithmetic is correct.

2.16 Common traps

Trap 1: confusing a sequence with a set

The ordered outcomes

$$(A, B, C) \quad \text{and} \quad (C, B, A)$$

are different sequences but the same set:

$$\{A, B, C\} = \{C, B, A\}.$$

Use sequences when order is recorded. Use sets when only membership is recorded.

Trap 2: treating indistinguishable objects as distinguishable

The letters in BANANA are not six distinct letters. The three A's are indistinguishable, and the two N's are indistinguishable. Therefore $6!$ overcounts the number of visible arrangements.

Trap 3: using combinations when roles matter

A committee is a set. A medal assignment is an ordered assignment. A group of three athletes and a gold-silver-bronze result are not the same kind of outcome.

Trap 4: using a representation that records too much

Sometimes an auxiliary representation is more detailed than the intended outcome. For example, ordered lists may be useful for deriving the number of committees, but the committee itself is not an ordered list. If the extra descriptions are not corrected for, the count is too large.

Trap 5: using a representation that records too little

Sometimes a representation hides distinctions that the model requires. If positions, roles, labels, or order are part of the outcome, then a representation that ignores them gives too small a count. A good representation must preserve exactly the distinctions that the recording rule declares visible.

2.17 What combinatorics teaches here

The role of this chapter is not to memorize more formulas. Its role is to teach how to build and count finite spaces of possibilities. A formula is useful only after the underlying structure has been recognized.

The general workflow is

situation $\longrightarrow$ recording rule $\longrightarrow$ elementary outcome $\longrightarrow$ space of possibilities $\longrightarrow$ count.

Each arrow in this workflow represents a mathematical decision. The same everyday description may lead to different counts if the recording rule changes. The same physical objects may be treated as distinguishable or indistinguishable depending on what is visible in the model. The same collection may be represented as a list, a set, an arrangement, or a count vector.

Good counting therefore has two parts. First, one must construct the correct model. Second, one must justify the count by a construction argument, an overcounting correction, or a representation such as a table, a tree, or a stars-and-bars diagram.

2.18 Final checklist

When solving a counting problem, write down the following before doing arithmetic:

1. What is the physical situation?
2. What information is recorded?
3. What is one elementary outcome?
4. What is the space of possibilities?
5. How can this space be represented: as a list, table, tree, set, tuple, arrangement, or count vector?
6. Are outcomes sequences, sets, arrangements, or count vectors?
7. Does the representation record exactly the distinctions required by the model?
8. Which construction, overcounting correction, or bijection gives the count?
9. What does the final number mean in the original situation?

This checklist is the main lesson of the chapter. The formulas are useful, but they are useful only when the underlying structure has been discovered first and the count has been justified.

Chapter 3

Discovering Formalism

3.1 Why formalism has to be discovered

The previous chapter taught a first lesson: before one counts, one must know what is being counted. The present chapter begins from a closely related but deeper observation. Before one assigns any numerical value to uncertainty, one must know what kind of object a statement about a situation selects.

Mathematics often begins when ordinary language becomes too elastic. In ordinary language we can say things such as

the result is favourable, at least one die shows 6,

the first toss is heads, it rains on Wednesday or Friday.

These statements are meaningful, but they are not yet mathematical objects. They have grammar, but not yet structure. To reason with them reliably, we must translate them into something that can be compared, combined, negated, represented, and eventually measured.

The central discovery of this chapter is that a statement about a situation acts as a test on elementary outcomes. An elementary outcome either satisfies the statement or it does not. Therefore a statement selects a set of elementary outcomes. This selected set is what probability theory calls an event.

Remark

The formal language is not introduced to make the subject look more abstract. It is introduced because without it we cannot reliably say what is being discussed.

At this stage, we are not primarily interested in numerical answers. We are interested in the birth of a language. We want to see how the following chain of ideas appears naturally:

situation $\longrightarrow$ elementary outcomes $\longrightarrow$ statements about outcomes $\longrightarrow$ sets of outcomes $\longrightarrow$ operations on sets

Only after this language exists can numerical probability be introduced responsibly.

3.2 A first universe of discourse

Every formal discussion needs a universe in which its statements are interpreted. In our setting, that universe is the set of all elementary outcomes under consideration.

Definition

The **sample space** Ω is the set of elementary outcomes that are possible under the recording rule chosen for the situation.

In this chapter, Ω should be read as a universe of discourse. It tells us what the word “outcome” means in the current model. A statement is not evaluated in the air. It is evaluated relative to Ω .

Example

Consider two coin tosses, with order recorded. The sample space is

$$\Omega = \{HH, HT, TH, TT\}.$$

The statement “exactly one head occurs” is true for HT and TH , but false for HH and TT . Thus the statement selects the set

$$\{HT, TH\}.$$

The selected set is the mathematical object corresponding to the statement.

The same phrase may select a different set if the sample space changes. For example, if only the number of heads is recorded, then the sample space is

$$\Omega' = \{0, 1, 2\}.$$

In that model, the statement “exactly one head occurs” selects

$$\{1\}.$$

The ordinary sentence is similar, but the formal object is different because the underlying space is different.

Remark

Formalism forces us to keep track of the level at which we are speaking. Are we describing the full ordered outcome, a set of visible outcomes, or only a numerical summary? The answer changes the mathematical object.

This is the methodological point of the chapter. We do not begin with symbols because symbols are fashionable. We begin with a question: what must be preserved so that reasoning remains honest? The symbols appear as a disciplined way to preserve exactly that structure.

3.3 Statements become events

Let Ω be fixed. A statement about the situation becomes mathematical when we can say exactly for which elementary outcomes it is true. The set of all such outcomes is called an event.

Definition

An **event** is a subset of the sample space Ω . If $A \subseteq \Omega$, then A consists of precisely those elementary outcomes for which a certain statement is true.

This definition may look simple, but it is a major conceptual move. It tells us that statements and sets can play the same role once Ω has been fixed:

$$\text{statement about the situation} \longleftrightarrow \text{subset of } \Omega.$$

The statement gives meaning. The subset gives structure.

Example

Suppose two dice are rolled and order is recorded. Then

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

The statement

the sum is 7

selects the event

$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

The statement

the first die is greater than the second

selects the event

$$B = \{(i, j) \in \Omega : i > j\}.$$

The notation $B = \{(i, j) \in \Omega : i > j\}$ is not a shortcut for avoiding thought. It records the rule by which membership in the event is decided.

3.4 Representing events as regions

A set of outcomes can be written as a list, but lists are not always the best representation. When Ω has a visible structure, an event should often be represented as a region inside that structure. This is not merely a visual aid. It is a way of seeing relations.

For two dice, the sample space is naturally represented as a 6×6 table:

	1	2	3	4	5	6
1	.	.	.	.	.	.
2	.	.	.	.	.	.
3	.	.	.	.	.	.
4	.	.	.	.	.	.
5	.	.	.	.	.	.
6	.	.	.	.	.	.

Each cell represents one ordered pair (i, j) . The row records the first die, and the column records the second die.

The event $B = \{(i, j) : i > j\}$ occupies the cells below the diagonal:

	1	2	3	4	5	6
1	.	.	.	.	.	.
2	X	.	.	.	.	.
3	X	X	.	.	.	.
4	X	X	X	.	.	.
5	X	X	X	X	.	.
6	X	X	X	X	X	.

The diagonal itself corresponds to $i = j$. The region above the diagonal corresponds to $i < j$. The geometry of the table reveals the structure of the statement.

Remark

A representation is good when it preserves the distinctions relevant to the statement and makes relations visible. A poor representation may hide the structure that the problem

is trying to teach.

3.5 Reading a region as a statement

The translation works in both directions. A verbal statement selects a set, but a marked set also asks to be interpreted as a statement.

Example

In the two-dice table, suppose the marked cells are exactly the diagonal:

	1	2	3	4	5	6
1	X	.	.	.	.	.
2	.	X	.	.	.	.
3	.	.	X	.	.	.
4	.	.	.	X	.	.
5	.	.	.	.	X	.
6	.	.	.	.	.	X

This region can be described in several equivalent ways:

the two dice show the same number,

the first result equals the second result,

$$\{(i, j) \in \Omega : i = j\}.$$

The ability to move between these descriptions is part of understanding the formalism.

Students often think that formal notation is a translation from language into symbols. That is only half of the work. One must also be able to read symbols and diagrams back into language. A set without interpretation is mute; a sentence without a set is vague. Mathematics becomes powerful when both sides illuminate each other.

3.6 Logical operations become set operations

Once statements have become events, logical operations on statements become set operations on events. This is one of the most important structural discoveries in elementary probability theory. The words

and, or, not

are not informal decorations. They determine precise operations on subsets of Ω .

Let $A \subseteq \Omega$ and $B \subseteq \Omega$ be events.

Language	Set operation	Meaning
A and B	$A \cap B$	outcomes satisfying both statements
A or B	$A \cup B$	outcomes satisfying at least one statement
not A	$A^c = \Omega \setminus A$	outcomes not satisfying A
A but not B	$A \setminus B$	outcomes satisfying A and not B

The table is not a dictionary of symbols to memorize. It expresses a deeper fact: logical structure and set structure are the same structure viewed in two languages.

Definition

For events $A, B \subseteq \Omega$:

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\},$$

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\},$$

$$A^c = \{\omega \in \Omega : \omega \notin A\}.$$

3.7 Compound statements in a two-dice universe

Consider again two dice, with

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

Define three events:

$$A = \{(i, j) : i + j = 7\},$$

$$B = \{(i, j) : i > j\},$$

$$C = \{(i, j) : i = 6 \text{ or } j = 6\}.$$

Now each compound statement can be built systematically.

The statement

the sum is 7 or at least one die shows 6

corresponds to

$$A \cup C.$$

The statement

the sum is 7 and at least one die shows 6

corresponds to

$$A \cap C.$$

The statement

the sum is 7 but the first die is not greater than the second

corresponds to

$$A \cap B^c.$$

The statement

at least one die shows 6 but the sum is not 7

corresponds to

$$C \setminus A = C \cap A^c.$$

This is the beginning of an algebra of events. We are not merely naming sets. We are learning to calculate with statements.

Remark

The purpose of the notation is not to replace thought. The purpose is to make the structure of thought inspectable. If a student writes $A \cap B^c$, they should also be able to say what it means in words and mark the corresponding region in the sample space.

3.8 Why “or” needs care

In everyday speech, the word “or” is sometimes ambiguous. In mathematics, the union $A \cup B$ means inclusive or: an outcome belongs to $A \cup B$ if it belongs to A , or to B , or to both.

This matters because the overlap is not automatically empty. If A and B overlap, then the union is not obtained by placing two disjoint blocks side by side. The intersection must be seen.

Example

Let

$$A = \{(i, j) : i + j = 7\}, \quad C = \{(i, j) : i = 6 \text{ or } j = 6\}.$$

The outcome $(1, 6)$ belongs to both A and C . The outcome $(6, 1)$ also belongs to both. Thus

$$A \cap C = \{(1, 6), (6, 1)\}.$$

The phrase “sum is 7 or at least one die shows 6” includes these outcomes once, not twice. The union is a set of outcomes, not a list of reasons.

This is a methodological lesson. A sentence may have several reasons for being true, but an elementary outcome either belongs to the event or it does not. Formalism prevents double counting at the level of meaning before any numerical computation begins.

3.9 Complementation and the boundary of the universe

The complement A^c is always taken relative to the current sample space Ω . This is why Ω must be fixed before complements are meaningful.

Example

If two dice are rolled and

$$A = \{(i, j) : i + j = 7\},$$

then

$$A^c = \{(i, j) \in \Omega : i + j \neq 7\}.$$

The complement is not an abstract collection of all things in the world for which the sum is not 7. It is the collection of outcomes inside the two-dice sample space for which the sum is not 7.

The complement teaches a philosophical lesson hidden inside a technical definition: negation is always negation within a universe. To say “not A ” mathematically, one must first say what counts as a possible case.

3.10 From events to observed frequencies

So far, an event is only a set of outcomes. This is already a substantial achievement: we can now state clearly what a sentence means. But probability theory eventually wants to do something more. It wants to attach numbers to events.

The first honest source of such numbers is observation. If an experiment is repeated many times, and we record how often each outcome or event occurs, then we obtain observed frequencies. These frequencies are not yet theoretical probabilities. They are empirical measurements. Nevertheless, they reveal the shape that a future theory of probability should have.

Suppose a die is rolled 1000 times. The elementary outcomes are

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Assume the recorded numbers are

$$n(\{1\}) = 168, \quad n(\{2\}) = 154, \quad n(\{3\}) = 181,$$

$$n(\{4\}) = 167, \quad n(\{5\}) = 160, \quad n(\{6\}) = 170.$$

For an event $A \subseteq \Omega$, define its observed frequency by

$$f(A) = \frac{n(A)}{1000},$$

where $n(A)$ is the number of recorded trials whose outcome belongs to A .

Example

Let

$$A = \{2, 4, 6\}.$$

Then A occurs whenever the recorded die result is 2, 4, or 6. Therefore

$$n(A) = n(\{2\}) + n(\{4\}) + n(\{6\}) = 154 + 167 + 170 = 491,$$

and hence

$$f(A) = \frac{491}{1000} = 0.491.$$

The addition is justified because the elementary outcomes 2, 4, and 6 are disjoint possibilities. A single roll cannot be both 2 and 4.

This example shows a transition. We did not begin with a formula for probability. We began with a set A , observed how many recorded outcomes fell inside that set, and divided by the total number of trials.

3.11 What frequencies teach us

Observed frequencies have structural properties. These properties are not arbitrary. They come from the way sets of outcomes are counted in repeated observations.

First, frequencies are non-negative:

$$f(A) \geq 0.$$

They cannot be negative because counts cannot be negative.

Second, frequencies are at most 1:

$$f(A) \leq 1.$$

The event A cannot occur more often than the total number of trials.

Third, the whole sample space occurs on every trial:

$$f(\Omega) = 1.$$

Every recorded outcome belongs to Ω by construction.

Fourth, the impossible event never occurs:

$$f(\emptyset) = 0.$$

No recorded outcome belongs to the empty set.

The most important structural property concerns disjoint events. If $A \cap B = \emptyset$, then no trial can contribute to both A and B . Therefore

$$n(A \cup B) = n(A) + n(B),$$

and after division by the total number of trials,

$$f(A \cup B) = f(A) + f(B).$$

Theorem

For observed frequencies on a fixed finite sample space:

$$f(\emptyset) = 0, \quad f(\Omega) = 1,$$

$$0 \leq f(A) \leq 1,$$

and if $A \cap B = \emptyset$, then

$$f(A \cup B) = f(A) + f(B).$$

These are not yet axioms. They are observations about how empirical frequencies behave. The axioms of probability will later abstract exactly this pattern.

3.12 When simple addition fails

If two events are not disjoint, simple addition counts the overlap twice. This is not a numerical accident; it is a structural fact about sets.

Let

$$M = \{1, 2, 3\}, \quad N = \{3, 4, 5\}.$$

Then

$$M \cap N = \{3\}.$$

If we add $n(M)$ and $n(N)$, the trials with outcome 3 are counted once as part of M and once as part of N . Therefore

$$n(M \cup N) = n(M) + n(N) - n(M \cap N).$$

At the frequency level,

$$f(M \cup N) = f(M) + f(N) - f(M \cap N).$$

Remark

The correction term is not a trick. It is the price of overlap. Whenever an outcome has two reasons to be included, addition counts it twice unless the overlap is subtracted once.

The philosophy remains the same: the numerical rule is not primary. The structure of the event comes first; the numerical rule follows from that structure.

3.13 From observed frequency to probability

Observed frequencies are tied to a particular record of trials. If the same experiment is repeated again, the frequencies may change slightly. Yet the structure suggested by frequencies is stable:

events receive numbers between 0 and 1, the whole sample space receives 1, the empty event receives 0, and disjoint unions receive sums.

This suggests a new mathematical object. Instead of recording what happened in one finite collection of trials, we introduce a function that assigns a number to each event in a way that preserves the structural rules discovered above.

Definition

A **probability assignment** on a finite sample space Ω is a function

$$P : \mathcal{P}(\Omega) \rightarrow [0, 1]$$

that assigns a number $P(A)$ to each event $A \subseteq \Omega$.

The notation $\mathcal{P}(\Omega)$ denotes the power set of Ω , that is, the set of all subsets of Ω . Since events are subsets of Ω , the domain of P is the collection of events.

But not every function from $\mathcal{P}(\Omega)$ to $[0, 1]$ deserves to be called probability. The function must respect the structure of events.

Theorem

In a finite setting, a probability assignment should satisfy:

1. **Non-negativity:** for every event A ,

$$P(A) \geq 0.$$

2. **Normalization:**

$$P(\Omega) = 1.$$

3. **Additivity for disjoint events:** if $A \cap B = \emptyset$, then

$$P(A \cup B) = P(A) + P(B).$$

These properties are not decorative. They are the minimum conditions needed for the numerical language to remain faithful to the set language.

3.14 Consequences of the axioms in the finite case

Once the axioms are accepted, several familiar rules become consequences rather than assumptions.

First, the impossible event has probability zero. Since

$$\Omega = \Omega \cup \emptyset$$

and the union is disjoint, additivity gives

$$P(\Omega) = P(\Omega) + P(\emptyset).$$

Subtracting $P(\Omega)$ from both sides gives

$$P(\emptyset) = 0.$$

Second, complements add up to one. For any event A , the events A and A^c are disjoint and

$$A \cup A^c = \Omega.$$

Therefore

$$P(A) + P(A^c) = P(\Omega) = 1,$$

so

$$P(A^c) = 1 - P(A).$$

Third, if $A \subseteq B$, then $P(A) \leq P(B)$. Indeed, B can be decomposed as

$$B = A \cup (B \setminus A),$$

where the two parts are disjoint. Hence

$$P(B) = P(A) + P(B \setminus A).$$

Since probabilities are non-negative,

$$P(B \setminus A) \geq 0,$$

and therefore

$$P(B) \geq P(A).$$

Finally, for arbitrary events A and B , the union formula follows from splitting the union into disjoint pieces:

$$A \cup B = A \cup (B \setminus A).$$

Thus

$$P(A \cup B) = P(A) + P(B \setminus A).$$

But B itself decomposes as

$$B = (B \setminus A) \cup (A \cap B),$$

so

$$P(B) = P(B \setminus A) + P(A \cap B).$$

Therefore

$$P(B \setminus A) = P(B) - P(A \cap B),$$

and hence

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Remark

The formulas are not separate facts to memorize. They are consequences of preserving the structure of events. Once the event algebra is understood, the probability rules become almost inevitable.

3.15 The axiomatic point of view

The modern theory of probability is axiomatic. This does not mean that it ignores intuition. It means that it takes the stable part of intuition and turns it into a structure that can be used reliably.

In finite spaces, the essential ideas are already visible:

- events are subsets of a sample space;
- probability assigns numbers to events;
- the whole space has probability 1;

- disjoint alternatives add.

For general probability spaces, the same philosophy is retained, but one must be more careful. Instead of assigning probabilities to all subsets of Ω , one usually assigns probabilities to a chosen family of events $\mathcal{F}$. This family must be closed under the operations needed for logical reasoning: complements, countable unions, and therefore also countable intersections. Such a family is called a σ -algebra.

Definition

A **probability space** is a triple

$$(\Omega, \mathcal{F}, P),$$

where Ω is a sample space, $\mathcal{F}$ is a σ -algebra of events, and $P : \mathcal{F} \rightarrow [0, 1]$ is a probability measure satisfying:

1. $P(A) \geq 0$ for all $A \in \mathcal{F}$,
2. $P(\Omega) = 1$,
3. if $A_1, A_2, A_3, \dots$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The third condition is countable additivity. In finite examples it reduces to the additivity rules already seen. In infinite settings it becomes a deep structural requirement: it says that probability remains compatible with decompositions into countably many disjoint alternatives.

This is the philosophical endpoint of the chapter. We began with informal statements about outcomes. We discovered that statements select sets. We discovered that logical relations between statements become set operations. We observed that frequencies assign numbers to events in a structured way. Finally, we abstracted the stable part of that structure into axioms.

The beauty of the formalism is that it is not imposed from outside. It is discovered by asking, step by step, what must be preserved if reasoning under uncertainty is to become mathematics.

Chapter 4

Sample Spaces and Elementary Probability

This chapter develops several canonical examples of sample spaces and elementary probability from the ground up. The emphasis is on understanding the experiment, deciding what should count as an elementary outcome, constructing the sample space, defining events, and assigning probabilities only after the mathematical model has been made explicit.

Different representations of the same probabilistic structure should be used throughout the chapter: lists, tables, trees, diagrams, and symbolic notation. The goal is to show that a sample space is an abstract object produced by analysis and modelling, not a collection of observed data.

Chapter 5

Conditional Probability, Independence, and Bayes' Theorem

This chapter explores the algebra of events and the concept of conditional probability, which are fundamental to understanding how probabilities are updated with new information.

Chapter 6

Discrete Random Variables and Probability Laws

This chapter introduces random variables through the discrete case. A random variable is viewed as a function that extracts a numerical quantity from an elementary outcome and allows probabilistic phenomena to be studied through numerical characteristics.

The chapter develops discrete probability laws, probability mass functions, cumulative distribution functions, expectation, variance, moments, and the interpretation of these quantities. Classical discrete models are presented as mathematical descriptions of specific random mechanisms.

Chapter 7

Continuous Random Variables and Probability Laws

This chapter develops the continuous counterpart of discrete probability laws. The central conceptual change is that probability is no longer concentrated at individual points; instead, probabilities of intervals become the fundamental objects.

The chapter introduces probability density functions, cumulative distribution functions, integrals, expectation, variance, moments, quantiles, and the interpretation of probability as area under a density. Continuous models such as the uniform, exponential, normal, gamma, and beta distributions are treated as tools for modelling continuous random phenomena.

Chapter 8

Empirical Distributions and Descriptive Statistics

This chapter studies empirical distributions and descriptive statistics obtained from data. The concepts developed earlier for theoretical probability laws are now examined through observed datasets.

Frequency tables, histograms, empirical distribution functions, means, medians, variances, standard deviations, quantiles, and related descriptive measures are introduced as tools for describing and understanding data.

Chapter 9

Sampling and Distributions of Statistics

This chapter introduces the idea that statistics computed from samples are themselves random variables. Sample means, proportions, variances, medians, and other summaries acquire their own probability laws when sampling is repeated.

The purpose of the chapter is to prepare the transition from descriptive statistics to statistical inference through the study of sampling distributions.

Chapter 10

Limit Theorems and Statistical Inference

This chapter studies the regularities that emerge when sample sizes become large and experiments are repeated many times. The law of large numbers and the central limit theorem provide the mathematical foundation for statistical inference.

The chapter introduces normal approximations, standard errors, confidence intervals, and the basic principles of inference from data.